



13 July 2021



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term I Examination, Summer 2021 Semester

Course: CSE 106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Exam Time: 1 Hour and 10 Minutes
Submission Time: 10 Minutes
Total Time: 1 hour 20 minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. **Prepare** truth table for the following compound proposition: [CO1,C2, Mark: 8]
 $((\neg A \vee \neg C) \vee B) \rightarrow (A \wedge B)$.
2. Without truth table **verify** the following equivalence: [CO1,C3, Mark: 8]
 $\neg(p \rightarrow q) \equiv (p \wedge \neg q)$.
3. Let $P(x, y)$ be the statement $x + y = 0$, where $x, y \in \mathbb{Z}^+$. **Determine** the truth values of $\exists x \forall y P(x, y)$ and $\forall x \forall y P(x, y)$. [CO1,C2, Mark: 8]
4. Let $Dhaka(x)$, $Coronavirus(x)$ and $Vaccinated(x)$ be the statements "x lives in Dhaka city," "x is affected by coronavirus," and "x will be vaccinated," respectively. Suppose that the domain consists of all people in the world. **Express** each of these statements using quantifiers; logical connectives; and $Dhaka(x)$, $Coronavirus(x)$ and $Vaccinated(x)$. [CO1,C3, Mark: 8]
 - A. Everyone in Dhaka City will be vaccinated.
 - B. Some people in Dhaka city are affected by coronavirus.
5. Use direct proof technique to prove the following theorem: [CO1,C3, Mark: 6]
 "If n is a multiple of 3, then n^3 is also a multiple of 3."
6. Use proof by contraposition technique to prove the following theorem: [CO1,C3, Mark: 6]
 "If $n^2 - 1$ is an even integer, then n is an odd integer."
7. Use mathematical induction to prove the following equality, for all natural numbers $n \geq 2$: [CO1,C3, Mark: 6]

$$\sum_{t=1}^{n-1} t(t+1) = \frac{n(n-1)(n+1)}{3}$$



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Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. **Prepare** truth table for the following compound proposition: [CO1,C2, Mark: 8]
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 $\neg(p \rightarrow q) \equiv (p \wedge \neg q)$.
3. Let $P(x, y)$ be the statement $x + y = 0$, where $x, y \in \mathbb{Z}^+$. **Determine** the truth values of $\exists x \forall y P(x, y)$ and $\forall x \forall y P(x, y)$. [CO1,C2, Mark: 8]
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 - A. Everyone in Dhaka City will be vaccinated.
 - B. Some people in Dhaka city are affected by coronavirus.
5. Use direct proof technique to prove the following theorem: [CO1,C3, Mark: 6]
 "If n is a multiple of 3, then n^3 is also a multiple of 3."
6. Use proof by contraposition technique to prove the following theorem: [CO1,C3, Mark: 6]
 "If $n^2 - 1$ is an even integer, then n is an odd integer."
7. Use mathematical induction to prove the following equality, for all natural numbers $n \geq 2$: [CO1,C3, Mark: 6]

$$\sum_{t=1}^{n-1} t(t+1) = \frac{n(n-1)(n+1)}{3}$$

Write answer of each question and intermediate steps in the right margin.

1. **Prepare** truth table for the following compound proposition:
 $((\neg p \vee \neg q) \vee r) \rightarrow (p \wedge q).$ [CO1,C2,
Mark: 8]
2. Using a truth table **verify** the following equivalence:
 $((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r) \equiv T.$ [CO1,C3,
Mark: 8]
3. Let $P(x, y, z)$ be the statement $x + y + z = 3$, where $x, y, z \in \mathbf{Z}$. **Determine** the truth values of $P(-1, 2, 3)$ and $P(5, -2, 0)$. [CO1,C2,
Mark: 8]
4. **Determine** the negation of the following quantified statement:
 $\exists x \forall y \exists z ((x \leq y) \wedge (x \geq z)).$ [CO1,C3,
Mark: 8]
5. **Use** direct proof technique to prove the following theorem:
“If n is an odd integer, then $n^2 - 1$ is an even integer.” [CO1,C3,
Mark: 6]
6. **Use** proof by contraposition technique to prove the following theorem:
“If $n^3 + 1$ is an even integer, then n is an odd integer.” [CO1,C3,
Mark: 6]
7. **Use** mathematical induction to prove that, $2^{3n} - 1$ is divisible by 7, for all natural numbers N except zero. [CO1,C3,
Mark: 6]



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term I Examination, Spring 2023 Semester

CSE106 Discrete Mathematics, Section-07
Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department

50 (25 will be counted for final grading)
1 Hour and 10 Minutes

Course:
Instructor:
Full Marks:
Time:

Note: There are *SEVEN* questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. Prepare truth table for the following compound proposition?
$$[(A \wedge B) \wedge (A \rightarrow C) \wedge (B \rightarrow C)] \leftrightarrow C$$

[CO1,C2, Mark: 8]
2. Verify the following equivalence without using truth table?
$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

[CO1,C3, Mark: 8]
3. Let $P(x,y)$ be the statements $2x + y = 4$, where $x, y \in \mathbb{Z}^+$. Determine the truth values of $\exists x \exists y P(x,y)$ and $\forall x \exists y P(x,y)$?

[CO1,C2, Mark: 8]
4. Determine the negation of the following quantified statement?
$$\forall x \exists y \forall z ((x \leq y) \wedge (y > z))$$

[CO1,C Mark:]
5. Use the direct proof technique to prove the following theorem?
"If n is divisible by 5, then n^4 is also divisible by 5."
6. Use proof by contraposition technique to prove the following theorem?
"If $(n^2 - 1)$ is a odd integer, n is an even integer."
7. Use mathematical induction to prove the following equality, for all natural number $n \geq 2$?
$$\left(1 - \frac{1}{2^2}\right) \cdot \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$



EAST WEST UNIVERSITY
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B.Sc. in Computer Science and Engineering Program
Mid Term I Examination, Spring 2022 Semester

Course: CSE106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time 10 Minutes for uploading
Total Time 1 Hour 20 Minutes

Note: There are *SEVEN* questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. **Prepare** truth table for the following compound proposition: [CO1,C2, Mark: 8]

$$[(A \vee B) \wedge (A \rightarrow C) \wedge (B \rightarrow C)] \rightarrow C$$
2. **Verify** the following equivalence without a truth table? [CO1,C3, Mark: 8]

$$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow q \Rightarrow (q \vee r) = T$$
3. Let $P(x,y)$ be the statements $x - y = 1$, where $x, y \in \mathbb{Z}^+$. **Determine** the truth values of $\exists x \forall y P(x,y)$ and $\forall x \exists y P(x,y)$? [CO1,C2, Mark: 8]
4. Let $Dhaka(x)$, $Student(x)$, $Coronavirus(x)$ and $Vaccinated(x)$ be the statements “x lives in Dhaka city,” “x is a student of East West University,” “x is infected by coronavirus,” and “x will be vaccinated” respectively. Suppose that the domain consists of all people in the world. **Express** each of these statements using quantifiers, logical connectives; and $Dhaka(x)$, $Student(x)$, $Coronavirus(x)$ and $Vaccinated(x)$? [CO1,C3, Mark: 8]
 - A. All students of East West University will not be vaccinated.
 - B. Some students of East West University living in Dhaka city have not been infected by coronavirus.
5. **Use** the direct proof technique to prove the following theorem: [CO1,C3, Mark: 6]
 “If n is divisible by 5, then n^2 is also divisible by 5.”
6. **Use** proof by contraposition technique to prove the following theorem: [CO1,C3, Mark: 6]
 “If $(n^2 + 1)$ is a odd integer, n is an even integer.”
7. **Use** mathematical induction to prove the following equality, for all natural numbers $n \geq 2$: [CO1,C3, Mark: 6]

$$\left(1 - \frac{1}{2^2}\right) \cdot \left(1 - \frac{1}{3^2}\right) \cdots \cdots \cdots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. **Prepare** truth table for the following compound proposition: [CO1,C2,
Mark: 8]

$$((\neg A \vee \neg C) \vee B) \rightarrow (A \wedge B).$$
2. Without truth table **verify** the following equivalence: [CO1,C3,
Mark: 8]

$$\neg(p \rightarrow q) \equiv (p \wedge \neg q).$$
3. Let $P(x, y)$ be the statement $x + y = 0$, where $x, y \in \mathbf{Z}^+$. **Determine** the truth values of $\exists x \forall y P(x, y)$ and $\forall x \forall y P(x, y)$. [CO1,C2,
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 “If $n^2 - 1$ is an even integer, then n is an odd integer.”
7. **Use** mathematical induction to prove the following equality, for all natural numbers $n \geq 2$: [CO1,C3,
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EAST WEST UNIVERSITY
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B.Sc. in Computer Science and Engineering Program
Mid Term I Examination, Spring 2022 Semester

Course: CSE106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time 10 Minutes for uploading
Total Time 1 Hour 20 Minutes

Note: There are *SEVEN* questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. **Prepare** truth table for the following compound proposition: [CO1,C2, Mark: 8]

$$[(A \vee B) \wedge (A \rightarrow C) \wedge (B \rightarrow C)] \rightarrow C$$
2. **Verify** the following equivalence without a truth table? [CO1,C3, Mark: 8]

$$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow q \Rightarrow (q \vee r) = T$$
3. Let $P(x,y)$ be the statements $x - y = 1$, where $x, y \in \mathbb{Z}^+$. **Determine** the truth values of $\exists x \forall y P(x,y)$ and $\forall x \exists y P(x,y)$? [CO1,C2, Mark: 8]
4. Let $Dhaka(x)$, $Student(x)$, $Coronavirus(x)$ and $Vaccinated(x)$ be the statements "x lives in Dhaka city," "x is a student of East West University," "x is infected by coronavirus," and "x will be vaccinated" respectively. Suppose that the domain consists of all people in the world. **Express** each of these statements using quantifiers, logical connectives; and $Dhaka(x)$, $Student(x)$, $Coronavirus(x)$ and $Vaccinated(x)$? [CO1,C3, Mark: 8]
 - A. All students of East West University will not be vaccinated.
 - B. Some students of East West University living in Dhaka city have not been infected by coronavirus.
5. **Use** the direct proof technique to prove the following theorem: [CO1,C3, Mark: 6]
 "If n is divisible by 5, then n^2 is also divisible by 5."
6. **Use** proof by contraposition technique to prove the following theorem: [CO1,C3, Mark: 6]
 "If $(n^2 + 1)$ is a odd integer, n is an even integer."
7. **Use** mathematical induction to prove the following equality, for all natural numbers $n \geq 2$: [CO1,C3, Mark: 6]

$$\left(1 - \frac{1}{2^2}\right) \cdot \left(1 - \frac{1}{3^2}\right) \cdots \cdots \cdots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$



13 July 2021



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B.Sc. in Computer Science and Engineering Program
Mid Term I Examination, Summer 2021 Semester

Course: CSE 106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Exam Time: 1 Hour and 10 Minutes
Submission Time: 10 Minutes
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- 1. Prepare** truth table for the following compound proposition:
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- 2. Without** truth table **verify** the following equivalence:
 $\neg(p \rightarrow q) \equiv (p \wedge \neg q)$. [CO1,C3, Mark: 8]
- 3. Let** $P(x, y)$ be the statement $x + y = 0$, where $x, y \in \mathbb{Z}^+$. **Determine** the truth values of $\exists x \forall y P(x, y)$ and $\forall x \forall y P(x, y)$. [CO1,C2, Mark: 8]
- 4. Let** $Dhaka(x)$, $Coronavirus(x)$ and $Vaccinated(x)$ be the statements "x lives in Dhaka city," "x is affected by coronavirus," and "x will be vaccinated," respectively. Suppose that the domain consists of all people in the world. **Express** each of these statements using quantifiers; logical connectives; and $Dhaka(x)$, $Coronavirus(x)$ and $Vaccinated(x)$. [CO1,C3, Mark: 8]
 - A.** Everyone in Dhaka City will be vaccinated.
 - B.** Some people in Dhaka city are affected by coronavirus.
- 5. Use** direct proof technique to prove the following theorem:
 "If n is a multiple of 3, then n^3 is also a multiple of 3." [CO1,C3, Mark: 6]
- 6. Use** proof by contraposition technique to prove the following theorem:
 "If $n^2 - 1$ is an even integer, then n is an odd integer." [CO1,C3, Mark: 6]
- 7. Use** mathematical induction to prove the following equality, for all natural numbers $n \geq 2$: [CO1,C3, Mark: 6]

$$\sum_{t=1}^{n-1} t(t+1) = \frac{n(n-1)(n+1)}{3}$$



EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Mid Term I Examination, Spring 2022 Semester

Course:	CSE106 Discrete Mathematics, Section-4
Instructor:	Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks:	50 (25 will be counted for final grading)
Time:	1 Hour and 10 Minutes
Additional Time	10 Minutes for uploading
Total Time	1 Hour 20 Minutes

Note: There are *SEVEN* questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. Prepare truth table for the following compound proposition: [CO1,C2, Mark: 8]

$$[(A \vee B) \wedge (A \rightarrow C) \wedge (B \rightarrow C)] \rightarrow C$$
2. Verify the following equivalence without a truth table? [CO1,C3, Mark: 8]

$$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow q \Rightarrow (q \vee r) = T$$
3. Let $P(x,y)$ be the statements $x - y = 1$, where $x, y \in \mathbb{Z}^+$. Determine the truth values of $\exists x \forall y P(x,y)$ and $\forall x \exists y P(x,y)$? [CO1,C2, Mark: 8]
4. Let Dhaka(x), Student(x), Coronavirus(x) and Vaccinated(x) be the statements "x lives in Dhaka city," "x is a student of East West University," "x is infected by coronavirus," and "x will be vaccinated" respectively. Suppose that the domain consists of all people in the world. Express each of these statements using quantifiers, logical connectives; and Dhaka(x), Student(x), Coronavirus(x) and Vaccinated(x)? [CO1,C3, Mark: 8]
 - A. All students of East West University will not be vaccinated.
 - B. Some students of East West University living in Dhaka city have not been infected by coronavirus.
5. Use the direct proof technique to prove the following theorem: [CO1,C3, Mark: 6]
 "If n is divisible by 5, then n^3 is also divisible by 5."
6. Use proof by contraposition technique to prove the following theorem: [CO1,C3, Mark: 6]
 "If $(n^2 + 1)$ is a odd integer, n is an even integer."
7. Use mathematical induction to prove the following equality, for all natural numbers $n \geq 2$: [CO1,C3, Mark: 6]

$$\left(1 - \frac{1}{2^2}\right) \cdot \left(1 - \frac{1}{3^2}\right) \cdots \cdots \cdots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$

1. **Prepare** truth table for the following compound proposition: [CO1,C2,
Mark: 8]

$$((\neg p \vee \neg q) \vee r) \rightarrow (p \wedge q).$$
2. Using a truth table **verify** the following equivalence: [CO1,C3,
Mark: 8]

$$(((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)) \equiv T.$$
3. Let $P(x, y, z)$ be the statement $x + y + z = 3$, where $x, y, z \in \mathbf{Z}$. **Determine** the truth values of $P(-1, 2, 3)$ and $P(5, -2, 0)$. [CO1,C2,
Mark: 8]
4. **Determine** the negation of the following quantified statement: [CO1,C3,
Mark: 8]

$$\exists x \forall y \exists z ((x \leq y) \wedge (x \geq z)).$$
5. **Use** direct proof technique to prove the following theorem: [CO1,C3,
Mark: 6]
 “If n is an odd integer, then $n^2 - 1$ is an even integer.”
6. **Use** proof by contraposition technique to prove the following theorem: [CO1,C3,
Mark: 6]
 “If $n^3 + 1$ is an even integer, then n is an odd integer.”
7. **Use** mathematical induction to prove that, $2^{3n} - 1$ is divisible by 7, for all natural numbers N except zero. [CO1,C3,
Mark: 6]



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term I Examination, Spring 2023 Semester

March 7, 2023

Course:

Instructor:

Full Marks:

Time:

CSE106 Discrete Mathematics, Section-06

Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department

50 (25 will be counted for final grading)

1 Hour and 10 Minutes

Note: There are *SEVEN* questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. Prepare truth table for the following compound proposition:
$$[(A \vee B) \wedge (A \rightarrow C) \wedge (B \rightarrow C)] \rightarrow C$$
[CO1,C2,
Mark: 8]
2. Verify the following equivalence without a truth table?
$$[(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q] \vee q \equiv T$$
[CO1,C3,
Mark: 8]
3. Let $P(x,y)$ be the statements $x \div y = 1$, where $x, y \in \mathbb{Z}$. Determine the truth values of $\exists x \forall y P(x,y)$ and $\forall x \exists y P(x,y)$?
$$\frac{x}{y} = 1 \quad x = y$$
[CO1,C2,
Mark: 8]
4. Let $Dhaka(x)$, $Student(x)$, $Coronavirus(x)$ and $Vaccinated(x)$ be the statements "x lives in Dhaka city," "x is a student of East West University," "x is infected by coronavirus," and "x will be vaccinated" respectively. Suppose that the domain consists of all people in the world. Express each of these statements using quantifiers, logical connectives; and $Dhaka(x)$, $Student(x)$, $Coronavirus(x)$ and $Vaccinated(x)$?
A. All students of East West University will be vaccinated.
B. Some students of East West University living in Dhaka city have been infected by coronavirus.
[CO1,C3,
Mark: 8]
5. Use the direct proof technique to prove the following theorem:
"If n is a multiple of 7, then n^4 is also a multiple of 7."
[CO1,C3,
Mark: 6]
6. Use the proof by contraposition technique to prove the following theorem:
"If $(n^2 - 1)$ is an even integer, n is an odd integer."
[CO1,C3,
Mark: 6]
7. Use mathematical induction to prove the following equality, for all natural numbers $n \geq 2$:
$$\prod_{i=2}^n \left(1 - \frac{1}{i^2}\right) = \frac{n+1}{2n}$$
[CO1,C3,
Mark: 6]



Mid2 solution DSU - Google ড্রাইভ

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EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test II, Summer 2022 Semester

Course:

CSE 106 Discrete Mathematics, Section-06

Instructor:

Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department

Marks:

10 (10 will be counted for final grading)

Exam Time:

15 Minutes

Answer ALL of them.

Let $S = \{0, 2, 4, 6\}$ and $T = \{1, 2, 3, 5\}$. Determine whether each of the following sets of ordered pairs is a function with domain S and codomain T . Also, it is one-to-one? Is it onto?

- 1. $\{(0, 2), (2, 4), (4, 6), (6, 0)\}$ not one to one and onto
- 2. $\{(2, 3), (4, 7), (0, 1), (6, 5)\}$ not one to one and onto
- 3. $\{(6, 3), (2, 1), (0, 3), (4, 5)\}$ not one to one and not onto

For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, and whether it is antisymmetric?

- a. $\{(2, 4), (4, 2), (3, 4), (4, 3)\}$ reflexive
- b. $\{(1, 2), (2, 3), (3, 4), (3, 3)\}$
- c. $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

(a) reflexive because first

Mid Term II Examination, Fall 2021 Semester

Course:	CSE106 Discrete Mathematics, Section-5
Instructor:	Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks:	50 (25 will be counted for final grading)
Time:	1 Hour and 10 Minutes
Additional Time	10 Minutes for uploading
Total Time	1 Hour 20 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. **Show** that, $(A \cup B) \cap (\bar{A} \cup B) = B$, by using the membership table? [CO4,C2, Mark: 8]
2. **Determine** whether the function $f(x) = x^n - 2x$, $n > 0$ from $\mathbf{R}$ to $\mathbf{R}$ is a one-to-one function. [CO4,C2, Mark: 8]
3. **Give** a recursive definition of the Sequence is given bellow: (by following the proper procedure) [CO4,C3, Mark: 8]
1, 1, 4, 13, 43, 142, 469.....
4. The following is a relation on $\{1, 2, 3, 4\}$. **Determine** whether the relation is symmetric by using matrix? [CO4,C3, Mark: 8]
$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3,4), (4, 3), (4, 4)\}$$
5. **How** many possible outcomes (head or tail) are there if you flip a coin five times? [CO2,C2, Mark: 6]
6. Each user on a computer system has a password, which is five to eight characters long, where each character is an uppercase letter or a digit or a special symbol (special symbols are @, & and %). Each password must contain at least one digit. **How** many possible passwords are there? [CO2,C3, Mark: 6]
7. **What** is the minimum required number of students in a class to guarantee that at least ten will get the same grade, if there are four possible grades, A, B, C, D? [CO2,C3, Mark: 6]



Cse106(sec 04)

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KB/s

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CSE106_DSU...



August 17, 2021



EAST WEST UNIVERSITY

Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term II Examination, Summer 2021 Semester

Course: CSE106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
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Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. $U = \{5, 7, x, y, 10, z\}$ is the universal set and $A = \{5, 10, z\}$, $B = \{7, x, 10, z\}$ are two sets. **Represent** these sets using binary string. Using binary string and bit-wise logic operation **determine** $A - B'$. [CO4,C2, Mark: 8]
2. **Determine** whether the function $f(x) = x^5 + 1$ from N to Z is a bijective function. [CO4,C2, Mark: 8]
3. **Determine** the $(fog)^{-1}(x)$, where $f(x) = x + 3$ and $g(x) = 5x - 2$? [CO4,C3, Mark: 8]
4. The following is a relation on $\{10, 20, 30, 40\}$. **Determine** whether the relation is transitive or not by using constructing relational matrix? [CO4,C3, Mark: 8]
 $R = \{(10, 10), (10, 20), (20, 10), (20, 20), (30, 30), (30, 40), (40, 40)\}$
5. **Determine** how many bit strings of length 6 can be formed, where the last bit is 0 and two consecutive 0s are not allowed. [CO2,C2, Mark: 6]
6. The name of a variable in the C programming language is a string that can contain uppercase letters, lowercase letters, digits, or underscores. Further, the first character in the string must be a letter, either uppercase or lowercase, or an underscore. If a variable can be 1 to 5 characters long, **determine** the number of different variables that can be named in C. [CO2,C3, Mark: 6]
7. Given 13 Pascal books, 15 Python books, 20 C++ books, 25 Java books, FORTRAN books and 35 COBOL books, **determine** the minimum number of books we must select to ensure that there are at least 3 books dealing with same programming language. [CO2,C3, Mark: 6]

More

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ID:



EAST WEST UNIVERSITY

Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Class Test II, Spring 2022 Semester

10

Course:	CSE 106 Discrete Mathematics, Section-4
Instructor:	Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks:	10 (10 will be counted for final grading)
Exam Time:	20 Minutes

Note: Answer ALL of them.

1. Let $A = \{2, 4, 8, 10\}$ and $B = \{w, x, y, z\}$. Which of the following sets of ordered pairs represent a one to one function? [Mark:6]

- a. $\{(2, w), (2, x), (2, y), (2, z)\} \Rightarrow$ ~~not~~ one to one
- b. $\{(4, w), (2, x), (10, z), (8, y)\} \Rightarrow$ one to one
- c. $\{(4, w), (2, x), (8, x), (10, y)\} \Rightarrow$ ~~not~~ one to one

2. Check whether the function, $f(x) = |x| + 4$ from Z to Z , is a one to one function or not? [Mark: 4]

Soln
let, $x_1, x_2 \in Z$

and $x_1 = 1$

$x_2 = -1$

when $x_1 = 1$

$$\therefore f(x_1) = |1| + 4 = 1 + 4 = 5$$

when $x_2 = -1$

$$\therefore f(x_2) = |-1| + 4 = 1 + 4 = 5$$

Hence, $f(x_1) = f(x_2)$

$\therefore f(x)$ is not one to one

CSE106 Discrete Mathematics, Section-07
Mid Term II Examination, Spring 2023 Semester

Course:
Instructor:
Full Marks:
Time:

CSE106 Discrete Mathematics, Section-07
Dr. Mohammad Salah Uddin, Associate Professor, CSE Department
50 (25 will be counted for final grading)
1 Hour and 10 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. If A and B are sets, then prepare the membership table for the following set? [CO4,C2, Mark: 8]
$$(A \cup B) \cap (\bar{A} \cup B) \cup U$$
2. Determine whether the function $f(x) = |x| + 3$ from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ is a bijective function or not? [CO4,C2, Mark: 8]
3. Give a recursive definition of the Sequence given below: (by following the proper procedure) [CO4,C3, Mark: 8]
 $1, 1, 12, 78, 540, 3708, 25488, \dots$
4. The following is a relation on $\{4, 6, 8\}$. Determine whether the relation is anti-symmetric? (by generating a relational matrix) [CO4,C3, Mark: 8]
 $R = \{(4, 4), (6, 6), (8, 8), (4, 6), (6, 4), (6, 8), (8, 6), (8, 4)\}$
5. Determine how many bit strings are there of length eight that do not have 0 in them? [CO2,C2, Mark: 6]
6. Jenny has nine different skirts, six different tops, ten different pairs of shoes, two different necklaces and five different bracelets. Choosing one of each item, in how many ways can Jenny dress up? [CO2,C3, Mark: 6]
7. A box contains 13 apples, 9 mangos, and 8 olives. Determine the number of random picks of fruits that must be taken to guarantee that you have had at least 3 apples? [CO2,C3, Mark: 6]

April 5, 2023



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term II Examination, Spring 2023 Semester

Course: CSE106 Discrete Mathematics, Section-07
Instructor: Dr. Mohammad Salah Uddin, Associate Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. If A and B are sets, then **prepare** the membership table for the following set? [CO4,C2, Mark: 8]
$$(A \cup B) \cap (\bar{A} \cap B) \cap U$$
2. **Determine** whether the function $f(x) = |2x|$ from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ is a bijective function or not? [CO4,C2, Mark: 8]
3. **Give** a recursive definition of the Sequence given below: (by following the proper procedure) [CO4,C3, Mark: 8]
$$1, 1, 12, 78, 540, 3708, 25488, \dots$$
4. The following is a relation on $\{4, 6, 8\}$. **Determine** whether the relation is anti-symmetric. (by generating a relational matrix) [CO4,C3, Mark: 8]
$$R = \{(4, 4), (6, 6), (8, 8), (4, 6), (6, 4), (6, 8), (8, 6)\}$$
5. **Determine** how many bit strings are there of length six that do not have 1 in them? [CO2,C2, Mark: 6]
6. Jenny has nine different skirts, six different tops, nine different pairs of shoes, two different necklaces and five different bracelets. Choosing one of each item, in **how** many ways can Jenny dress up? [CO2,C3, Mark: 6]
7. A box contains 13 apples, 9 mangoes, and 8 olives. **Determine** the number of random picks of fruits that must be taken to guarantee that you have had at least 3 mangos? [CO2,C3, Mark: 6]



Cse106(sec 04)

1
KB/s

43% 11:55 am



CSE106_DSU...



August 17, 2021



EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Mid Term II Examination, Summer 2021 Semester

Course: CSE106 Discrete Mathematics, Section-4
 Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
 Full Marks: 50 (25 will be counted for final grading)
 Time: 1 Hour and 10 Minutes
 Additional Time: 10 Minutes for uploading
 Total Time: 1 Hour 20 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. $U = \{5, 7, x, y, 10, z\}$ is the universal set and $A = \{5, 10, z\}$, $B = \{7, x, 10, z\}$ are two sets. **Represent** these sets using binary string. Using binary string and bit-wise logic operation **determine** $A - B'$. [CO4,C2, Mark: 8]
2. **Determine** whether the function $f(x) = x^5 + 1$ from N to Z is a bijective function. [CO4,C2, Mark: 8]
3. **Determine** the $(fog)^{-1}(x)$, where $f(x) = x + 3$ and $g(x) = 5x - 2$? [CO4,C3, Mark: 8]
4. The following is a relation on $\{10, 20, 30, 40\}$. **Determine** whether the relation is transitive or not by using constructing relational matrix? [CO4,C3, Mark: 8]
 $R = \{(10, 10), (10, 20), (20, 10), (20, 20), (30, 30), (30, 40), (40, 40)\}$
5. **Determine** how many bit strings of length 6 can be formed, where the last bit is 0 and two consecutive 0s are not allowed. [CO2,C2, Mark: 6]
6. The name of a variable in the C programming language is a string that can contain uppercase letters, lowercase letters, digits, or underscores. Further, the first character in the string must be a letter, either uppercase or lowercase, or an underscore. If a variable can be 1 to 5 characters long, **determine** the number of different variables that can be named in C. [CO2,C3, Mark: 6]
7. Given 13 Pascal books, 15 Python books, 20 C++ books, 25 Java books, FORTRAN books and 35 COBOL books, **determine** the minimum number of books we must select to ensure that there are at least 3 books dealing with same programming language. [CO2,C3, Mark: 6]

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Cse106(sec 04)

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CSE106_DSU...



August 17, 2021



EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Mid Term II Examination, Summer 2021 Semester

Course: CSE106 Discrete Mathematics, Section-4
 Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
 Full Marks: 50 (25 will be counted for final grading)
 Time: 1 Hour and 10 Minutes
 Additional Time: 10 Minutes for uploading
 Total Time: 1 Hour 20 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. $U = \{5, 7, x, y, 10, z\}$ is the universal set and $A = \{5, 10, z\}$, $B = \{7, x, 10, z\}$ are two sets. **Represent** these sets using binary string. Using binary string and bit-wise logic operation **determine** $A - B'$. [CO4,C2, Mark: 8]
2. **Determine** whether the function $f(x) = x^5 + 1$ from N to Z is a bijective function. [CO4,C2, Mark: 8]
3. **Determine** the $(fog)^{-1}(x)$, where $f(x) = x + 3$ and $g(x) = 5x - 2$? [CO4,C3, Mark: 8]
4. The following is a relation on $\{10, 20, 30, 40\}$. **Determine** whether the relation is transitive or not by using constructing relational matrix? [CO4,C3, Mark: 8]
 $R = \{(10, 10), (10, 20), (20, 10), (20, 20), (30, 30), (30, 40), (40, 40)\}$
5. **Determine** how many bit strings of length 6 can be formed, where the last bit is 0 and two consecutive 0s are not allowed. [CO2,C2, Mark: 6]
6. The name of a variable in the C programming language is a string that can contain uppercase letters, lowercase letters, digits, or underscores. Further, the first character in the string must be a letter, either uppercase or lowercase, or an underscore. If a variable can be 1 to 5 characters long, **determine** the number of different variables that can be named in C. [CO2,C3, Mark: 6]
7. Given 13 Pascal books, 15 Python books, 20 C++ books, 25 Java books, FORTRAN books and 35 COBOL books, **determine** the minimum number of books we must select to ensure that there are at least 3 books dealing with same programming language. [CO2,C3, Mark: 6]

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EAST WEST UNIVERSITY

Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term II Examination, Fall 2020 Semester

Course: CSE106 Discrete Mathematics, Section-7
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time: 10 Minutes for uploading
Total Time: 1 Hour 20 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

$U = \{x, y, z, 1, 2, 3\}$ is the universal set and $A = \{2, x, z\}$, and $B = \{3, y, z\}$ are two sets. **Represent** these sets using binary string. **Determine** $A \cap B$ using binary string and bit-wise logic operation. [C] [M]

Determine whether the function $f(x) = x^2 - x$, $n > 0$ from $\mathbb{Z}^+$ to $\mathbb{Z}$ is a one-to-one function. [C] [M]

A recursive function $f(x)$ from nonnegative integer to nonnegative integer is defined as follows: [C] [M]

$$f(0) = 1, \quad f(1) = 1, \quad f(x+1) = f(x)f(x-1) \\ \text{for } x = 1, 2, 3, 4, \dots$$

Find the value of $f(6)$.

The following is a relation on $\{1, 2, 3\}$. **Determine** whether the relation is symmetric. [C] [M]

$$R = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2), (3, 3)\}$$

Calculate the number of ways an 8-bit string can be formed that either starts with 10 or ends with 11. [C] [M]

Suppose that a password for a computer system must have at least 7, but no more than 9 characters, where each character in the password is a lowercase English letter, an uppercase English letter, a digit, or one of the five special characters %, >, <, !, and =. **Compute**, how many different passwords are available for this computer system containing at least one occurrence of at least one of the special characters? [C] [M]

A bowl contains 10 red balls and 10 blue balls; a woman selects ball at random without looking at them. **Compute**, how many balls must she select to be sure of having at least three blue balls? [C] [M]

Mid Term II Examination, Fall 2021 Semester

Course:	CSE106 Discrete Mathematics, Section-5
Instructor:	Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks:	50 (25 will be counted for final grading)
Time:	1 Hour and 10 Minutes
Additional Time	10 Minutes for uploading
Total Time	1 Hour 20 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. **Show** that, $(A \cup B) \cap (\bar{A} \cup B) = B$, by using the membership table? [CO4,C2, Mark: 8]
2. **Determine** whether the function $f(x) = x^n - 2x$, $n > 0$ from $\mathbf{R}$ to $\mathbf{R}$ is a one-to-one function. [CO4,C2, Mark: 8]
3. **Give** a recursive definition of the Sequence is given bellow: (by following the proper procedure) [CO4,C3, Mark: 8]
1, 1, 4, 13, 43, 142, 469.....
4. The following is a relation on $\{1, 2, 3, 4\}$. **Determine** whether the relation is symmetric by using matrix? [CO4,C3, Mark: 8]
 $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}$
5. **How** many possible outcomes (head or tail) are there if you flip a coin five times? [CO2,C2, Mark: 6]
6. Each user on a computer system has a password, which is five to eight characters long, where each character is an uppercase letter or a digit or a special symbol (special symbols are @, & and %). Each password must contain at least one digit. **How** many possible passwords are there? [CO2,C3, Mark: 6]
7. **What** is the minimum required number of students in a class to guarantee that at least ten will get the same grade, if there are four possible grades, A, B, C, D? [CO2,C3, Mark: 6]



December 7, 2020

**EAST WEST UNIVERSITY****Department of Computer Science and Engineering****B.Sc. in Computer Science and Engineering Program****Mid Term II Examination, Fall 2020 Semester**

Course: CSE106 Discrete Mathematics, Section-8
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time: 10 Minutes for uploading
Total Time: 1 Hour 20 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. **Show** that, $(A \cap B) \cup (\bar{A} \cap B) = B$, by using membership table? [CO4,C2, Mark: 8]
2. **Determine** whether the function $f(x) = |x|$ from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ is a one-to-one function. [CO4,C2, Mark: 8]
3. **Give** a recursive definition of the Sequence is given bellow. (by following the proper procedure) [CO4,C3, Mark: 8]
 $-1, 2, -1, 5, 2, 17, 23, \dots$
4. The following is a relation on $\{1, 2, 3, 4\}$. **Determine** whether the relation is transitive. [CO4,C3, Mark: 8]
 $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}$
5. **How** many strings are there of length seven lowercase letters that have the letter lower case x in them? [CO2,C2, Mark: 6]
6. Each user on a computer system has a password, which is seven to nine characters long, where each character is an uppercase letter or a digit or a special symbol (special symbols are +, @, & and %). Each password must contain at least one digit and one special symbol. **How** many possible passwords are there? [CO2,C3, Mark: 6]
7. Suppose there are 5 different types of ice cream you like. **How** many random samples ice cream must be eaten to guarantee that you have had at least 7 samples of one type? [CO2,C3, Mark: 6]

Mid Term II Examination, Fall 2021 Semester

Course:	CSE106 Discrete Mathematics, Section-5
Instructor:	Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks:	50 (25 will be counted for final grading)
Time:	1 Hour and 10 Minutes
Additional Time	10 Minutes for uploading
Total Time	1 Hour 20 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. **Show that, $(A \cup B) \cap (\bar{A} \cup B) = B$, by using the membership table?** [CO4,C2, Mark: 8]
2. **Determine** whether the function $f(x) = x^n - 2x$, $n > 0$ from **R** to **R** is a one-to-one function. [CO4,C2, Mark: 8]
3. **Give a recursive definition of the Sequence** is given bellow: (by following the proper procedure) [CO4,C3, Mark: 8]
1, 1, 4, 13, 43, 142, 469.....
4. The following is a relation on $\{1, 2, 3, 4\}$. **Determine** whether the relation is symmetric by using matrix? [CO4,C3, Mark: 8]
 $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3,4), (4, 3), (4, 4)\}$
5. **How** many possible outcomes (head or tail) are there if you flip a coin five times? [CO2,C2, Mark: 6]
6. Each user on a computer system has a password, which is five to eight characters long, where each character is an uppercase letter or a digit or a special symbol (special symbols are @, & and %). Each password must contain at least one digit. **How** many possible passwords are there? [CO2,C3, Mark: 6]
7. **What** is the minimum required number of students in a class to guarantee that at least ten will get the same grade, if there are four possible grades, A, B, C, D? [CO2,C3, Mark: 6]



Find the value of $f(5)$.

[CO4, C2,
Mark: 8]

The following set is a relation on the set $\{1, 2, 3, 4\}$. Determine whether the relation is symmetric.

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

[CO2, C2,
Mark: 6]

Calculate the number of ways an 8-bit string can be formed that either starts with 10 or ends with 101.

[CO2, C3,
Mark: 6]

Determine how many bit strings of length 5 can be formed, where consecutive 101 are not allowed.

[CO2, C3,
Mark: 6]

A boy has 20 red balls, 30 blue balls, 40 yellow balls, and 50 pink balls. He selects balls at random without looking at them. Calculate the minimum number of balls he must select to be sure that at least 4 balls are of the same color.



Ca

April 5, 2023



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term II Examination, Spring 2023 Semester

Course: CSE106 Discrete Mathematics, Section-07
Instructor: Dr. Mohammad Salah Uddin, Associate Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. If A and B are sets, then **prepare** the membership table for the following set? [CO4,C2, Mark: 8]

$$(A \cup B) \cap (\bar{A} \cap B) \cap U$$

2. **Determine** whether the function $f(x) = |2x|$ from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ is a bijective function or not? [CO4,C2, Mark: 8]

3. **Give** a recursive definition of the Sequence given below: (by following the proper procedure) [CO4,C3, Mark: 8]

1, 1, 12, 78, 540, 3708, 25488.....

4. The following is a relation on $\{4, 6, 8\}$. **Determine** whether the relation is anti-symmetric. (by generating a relational matrix) [CO4,C3, Mark: 8]

$$R = \{(4, 4), (6, 6), (8, 8), (4, 6), (6, 4), (6, 8), (8, 6)\}$$

5. **Determine** how many bit strings are there of length six that do not have 1 in them? [CO2,C2, Mark: 6]

6. Jenny has nine different skirts, six different tops, nine different pairs of shoes, two different necklaces and five different bracelets. Choosing one of each item, in how many ways can Jenny dress up? [CO2,C3, Mark: 6]

7. A box contains 13 apples, 9 mangoes, and 8 olives. **Determine** the number of random picks of fruits that must be taken to guarantee that you have had at least 3 mangos? [CO2,C3, Mark: 6]



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term II Examination, Spring 2023 Semester

Course:
Instructor:
Full Marks:
Time:

CSE106 Discrete Mathematics, Section-07
Dr. Mohammad Salah Uddin, Associate Professor, CSE Department
50 (25 will be counted for final grading)
1 Hour and 10 Minutes

April 5, 2023

Nehidul Islam

2022-3-60-072

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. If A and B are sets, then prepare the membership table for the following set? [CO4,C2, Mark: 8]
$$(A \cup B) \cap (\bar{A} \cup B) \cup U$$
2. Determine whether the function $f(x) = |x| + 3$ from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ is a bijective function or not? [CO4,C2, Mark: 8]
3. Give a recursive definition of the Sequence given below: (by following the proper procedure) [CO4,C3, Mark: 8]
 $1, 1, 12, 78, 540, 3708, 25488, \dots$
4. The following is a relation on $\{4, 6, 8\}$. Determine whether the relation is anti-symmetric? (by generating a relational matrix) [CO4,C3, Mark: 8]
 $R = \{(4, 4), (6, 6), (8, 8), (4, 6), (6, 4), (6, 8), (8, 6), (8, 4)\}$
5. Determine how many bit strings are there of length eight that do not have 0 in them? [CO2,C2, Mark: 6]
6. Jenny has nine different skirts, six different tops, ten different pairs of shoes, two different necklaces and five different bracelets. Choosing one of each item, in how many ways can Jenny dress up? [CO2,C3, Mark: 6]
7. A box contains 13 apples, 9 mangos, and 8 olives. Determine the number of random picks of fruits that must be taken to guarantee that you have had at least 3 apples? [CO2,C3, Mark: 6]



Sultan Mahmud Shuvo

UNIVERSITY

Mid Term II Examination, Fall 2021 Semester

Course: CSE106 Discrete Mathematics, Section-5
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time 10 Minutes for uploading
Total Time 1 Hour 20 Minutes

Note: There are SEVEN questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

- Show** that, $(A \cup B) \cap (\bar{A} \cup B) = B$, by using the membership table? [CO4,C2, Mark: 8]
- Determine** whether the function $f(x) = x^n - 2x$, $n > 0$ from $\mathbf{R}$ to $\mathbf{R}$ is a one-to-one function. [CO4,C2, Mark: 8]
- Give** a recursive definition of the Sequence is given bellow: (by following the proper procedure) [CO4,C3, Mark: 8]
 1, 1, 4, 13, 43, 142, 469.....
- The following is a relation on $\{1, 2, 3, 4\}$. **Determine** whether the relation is symmetric by using matrix? [CO4,C3, Mark: 8]
 $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}$
- How** many possible outcomes (head or tail) are there if you flip a coin five times? [CO2,C2, Mark: 6]
- Each user on a computer system has a password, which is five to eight characters long, where each character is an uppercase letter or a digit or a special symbol (special symbols are @, & and %). Each password must contain at least one digit. **How** many possible passwords are there? [CO2,C3, Mark: 6]
- What** is the minimum required number of students in a class to guarantee that at least ten will get the same grade, if there are four possible grades, A, B, C, D? [CO2,C3, Mark: 6]



More



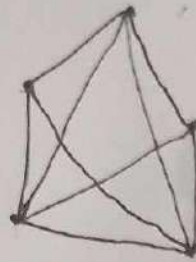
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Lecturer:
Full Marks:
Exam Time:

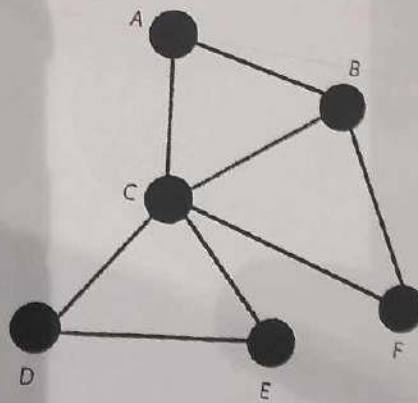
Science and Engineering
CSE 106 Discrete Mathematics, Section-06
Dr. Mohammad Salah Uddin, Associate Professor, CSE Department
10 (10 will be counted for final grading)
10 Minutes

Note: Answer ALL of them.

1. Draw a W_5 graph?



Determine whether the following graph is bipartite or not?

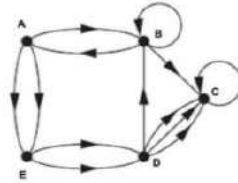


[Mark: 6]

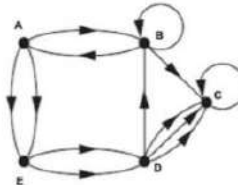
← CSE106_Final.pdf



Mark: 4]



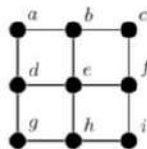
5. **Represent** the following graph using an adjacency matrix?

[CO4,C3,
Mark: 8]

Page 1 of 2

September 14, 2021

6. **Determine** whether the following graph is a bipartite graph or not?

[CO4,C3,
Mark: 6]

7. A full 4-ary tree has 3 internal vertices. **Determine** the number of vertices and number of leaves of that tree?

[CO4,C2,
Mark: 8]

2 / 2





September 14

1 / 2

**EAST WEST UNIVERSITY****Department of Computer Science and Engineering****B.Sc. in Computer Science and Engineering Program****Final Examination, Summer 2021 Semester**

Course: CSE106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time 10 Minutes for uploading
Total Time 1 Hour 20 Minutes

Note: There are *Seven* questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. Show that, $(x^3 + 2x)/(2x + 1)$ is $O(x^2)$? [CO3,C2, Mark: 8]

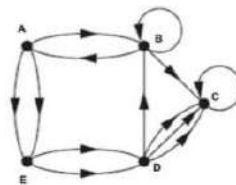
2. Determine the time complexity of the following algorithm as a function of n ? [CO3,C3, Mark: 8]

```

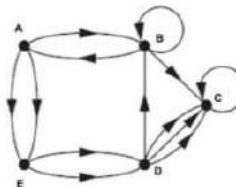
Input: A positive integer  $n$ .
Output: A integer (returned by the algorithm).
 $i = 1$ 
 $S = 0$ 
while ( $i \leq n^2$ ) do
    if  $i \% 2 \neq 0$  then  $S = S + i$ 
    if  $i \% 3 \neq 0$  then  $S = S - i$ 
     $i = i + 1$ 
return  $S$ 
  
```

3. Calculate $\gcd(910, 150)$ and $\text{lcm}(120, 240)$ using prime factorization technique? [CO3,C3, Mark: 8]

4. Determine in-degrees and out-degrees of all vertices of the following graph? [CO4,C2, Mark: 4]



5. Represent the following graph using an adjacency matrix? [CO4,C3, Mark: 8]



Note: There are *SEVEN* questions, answer ALL of them, Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. Show that the Big-O of, $f(x) = \frac{x^2 + 2x}{2x + 1}$ is x^2 ? [CO3,C2, Mark: 8]

2. Determine the time complexity of the following algorithm as a function of n? [CO3,C3, Mark: 8]

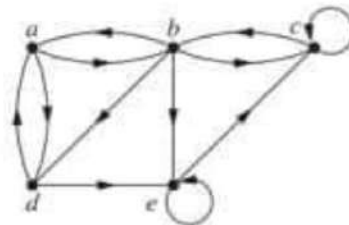
```

procedure matrix_man (n, resultn: integers)
  for i = 1 to n do
    for j = 1 to n do
      resulti = i
    end_for
  end_for
  return resulti

```

3. Calculate gcd(9910, 1550) and lcm(1120, 2240) using prime factorization technique? [CO3,C3, Mark: 8]

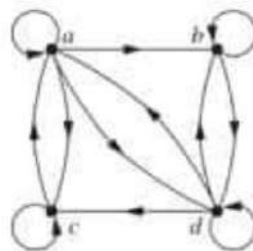
4. Determine in-degrees and out-degrees of all vertices of the following graph? [CO4,C2, Mark: 4]



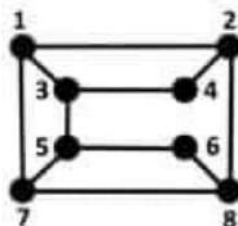
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5. Represent the following graph using an adjacency matrix? [CO4,C3, Mark: 8]



6. Determine whether the following graph is a bipartite graph or not? [CO4,C3, Mark: 6]



7. A full 4-ary tree has 40 leaves. Determine the number of vertices and internal vertices of that tree? [CO4,C2, Mark: 8]



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Final Examination, Fall 2020 Semester

Course: CSE106 Discrete Mathematics, Section-8
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time: 10 Minutes for uploading
Total Time: 1 Hour 20 Minutes

1/2



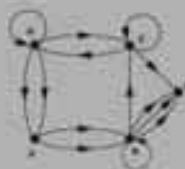
Note: There are Seven questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. Show that, $(x^3 + 2x)/(2x + 1) \text{ is } O(x^2)$. [CO3,C2, Mark: 8]
2. Determine the time complexity of the following algorithm. [CO3,C3, Mark: 8]

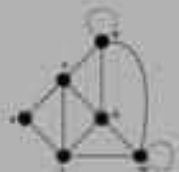
```

Input: A positive integer n.
Output: A integer (returned by the algorithm).
i ← 1
j ← 0
while (i < 2n) do
    i ← i + 1
return j
    
```

3. Calculate $\text{gcd}(910, 150)$ and $\text{lcm}(120, 240)$ using prime factorization technique. [CO3,C3, Mark: 8]
4. Determine in-degrees and out-degrees of all vertices of the following graph. [CO4,C2, Mark: 4]



5. Represent the following graph using an incidence matrix. [CO4,C3, Mark: 8]



6. Determine whether the following graph is a bipartite graph or not. [CO4,C3, Mark: 6]





Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Final Examination, Spring 2021 Semester

Course: CSE106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 68 (34 will be counted for final grading)
Time: 1 Hour and 20 Minutes
Additional Time: 10 Minutes for uploading
Total Time: 1 Hour 30 Minutes

Note: There are *Ten* questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. Show that, $(x^3 + 2x)/(2x + 1)$ is $O(x^2)$. [CO3,C2, Mark: 8]

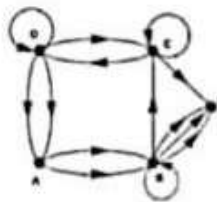
2. Determine the time complexity of the following algorithm. [CO3,C3, Mark: 8]

```

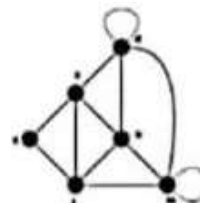
Input: A positive integer n.
Output: A integer (returned by the algorithm).
i = 1
S = 0
while (i < 3n) do
    i = i + 1
return S
  
```

3. Calculate gcd(910, 150) and lcm(120, 240) using prime factorization technique. [CO3,C3, Mark: 8]

4. Determine in-degrees and out-degrees of all vertices of the following graph. [CO4,C2, Mark: 4]



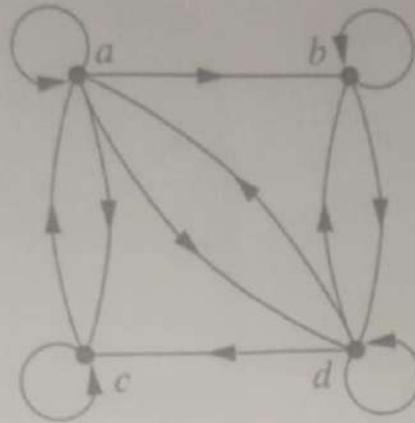
5. Represent the following graph using an incidence matrix. [CO4,C3, Mark: 8]



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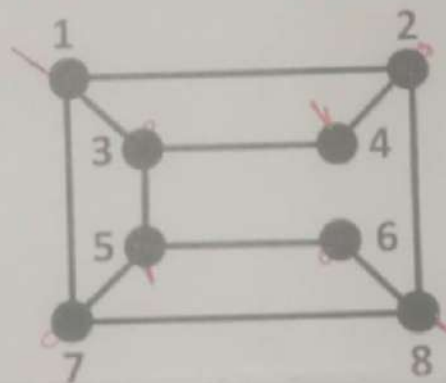
5. Represent the following graph using an adjacency matrix?

[CO4,C3,
Mark: 8]



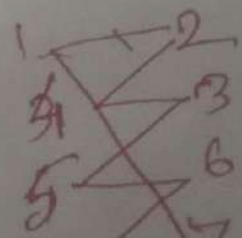
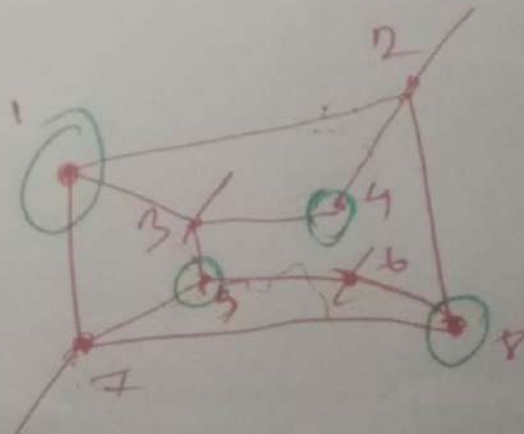
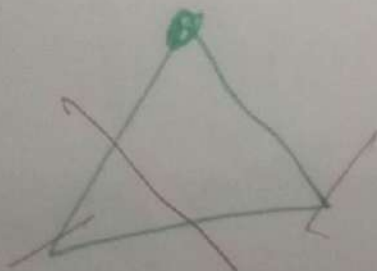
6. Determine whether the following graph is a bipartite graph or not?

[CO4,C3,
Mark: 6]



7. Draw a complete 2-ary tree of height 3?

[CO4,C2,
Mark: 8]



Course: CSE106 Discrete Mathematics, Section-7
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes

Note: There are *SEVEN* questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

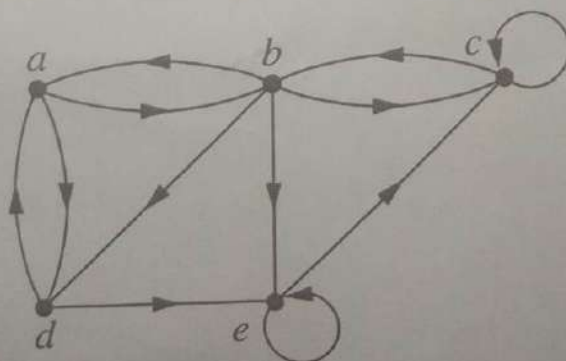
1. Show that the Big-O of, $f(x) = \frac{x^2 + 2x}{2x+1}$ is x^2 ? [CO3,C2, Mark: 8]
2. Determine the time complexity of the following algorithm as a function of n ? [CO3,C3, Mark: 8]

```

procedure matrix_man (n, resultnn: integers)
    for i = 1 to n do
        for j = 1 to n do
            resultij = i
        end_for
    end_for
    return resultij

```

3. Calculate gcd(9910, 1550) and lcm(1120, 2240) using prime factorization technique? [CO3,C3, Mark: 8]
4. Determine in-degrees and out-degrees of all vertices of the following graph? [CO4,C2, Mark: 4]



September 14

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**EAST WEST UNIVERSITY****Department of Computer Science and Engineering****B.Sc. in Computer Science and Engineering Program****Final Examination, Summer 2021 Semester**

Course: CSE106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time 10 Minutes for uploading
Total Time 1 Hour 20 Minutes

Note: There are *Seven* questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. Show that, $(x^3 + 2x)/(2x + 1)$ is $O(x^2)$? [CO3,C2, Mark: 8]

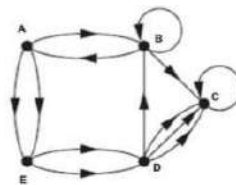
2. Determine the time complexity of the following algorithm as a function of n ? [CO3,C3, Mark: 8]

```

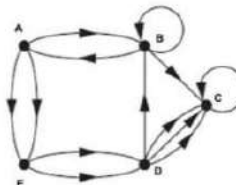
Input: A positive integer  $n$ .
Output: A integer (returned by the algorithm).
 $i = 1$ 
 $S = 0$ 
while ( $i \leq n^2$ ) do
    if  $i \% 2 \neq 0$  then  $S = S + i$ 
    if  $i \% 3 \neq 0$  then  $S = S - i$ 
     $i = i + 1$ 
return  $S$ 
  
```

3. Calculate $\gcd(910, 150)$ and $\text{lcm}(120, 240)$ using prime factorization technique? [CO3,C3, Mark: 8]

4. Determine in-degrees and out-degrees of all vertices of the following graph? [CO4,C2, Mark: 4]



5. Represent the following graph using an adjacency matrix? [CO4,C3, Mark: 8]



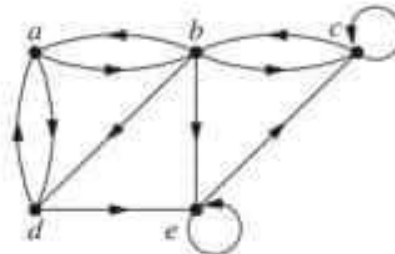
1. Show that the Big-O of, $f(x) = \frac{x^2 + 2x}{2x + 1}$ is x^2 ? [CO3,C2, Mark: 8]
2. Determine the time complexity of the following algorithm as a function of n? [CO3,C3, Mark: 8]

```

procedure matrix_man (n, resultmn: integers)
  for i = 1 to n do
    for j = 1 to n do
      resultij = i
    end_for
  end_for
  return resultij

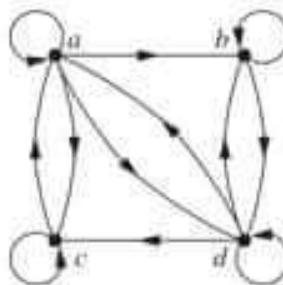
```

- | | | |
|----|--|-------------------|
| 3. | Calculate $\text{gcd}(9910, 1550)$ and $\text{lcm}(1120, 2240)$ using prime factorization technique? | [CO3,C3, Mark: 8] |
| 4. | Determine in-degrees and out-degrees of all vertices of the following graph? | [CO4,C2, Mark: 4] |

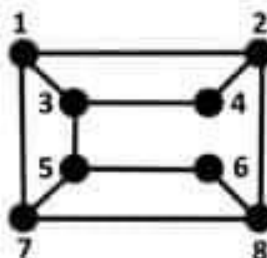


January 18, 2022

5. Represent the following graph using an adjacency matrix? [CO4,C3, Mark: 8]



6. Determine whether the following graph is a bipartite graph or not? [CO4,C3, Mark: 6]



7. A full 4-ary tree has 40 leaves. Determine the number of vertices and internal vertices of that tree? [CO4, C2, Mark: 8]



September 14

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**EAST WEST UNIVERSITY****Department of Computer Science and Engineering****B.Sc. in Computer Science and Engineering Program****Final Examination, Summer 2021 Semester**

Course: CSE106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time 10 Minutes for uploading
Total Time 1 Hour 20 Minutes

Note: There are *Seven* questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. Show that, $(x^3 + 2x)/(2x + 1)$ is $O(x^2)$? [CO3,C2, Mark: 8]

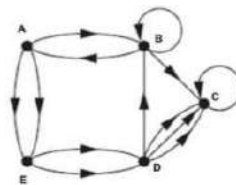
2. Determine the time complexity of the following algorithm as a function of n ? [CO3,C3, Mark: 8]

```

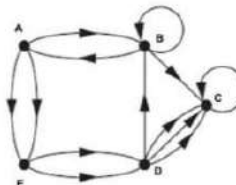
Input: A positive integer  $n$ .
Output: A integer (returned by the algorithm).
 $i = 1$ 
 $S = 0$ 
while ( $i \leq n^2$ ) do
    if  $i \% 2 \neq 0$  then  $S = S + i$ 
    if  $i \% 3 \neq 0$  then  $S = S - i$ 
     $i = i + 1$ 
return  $S$ 
  
```

3. Calculate $\gcd(910, 150)$ and $\text{lcm}(120, 240)$ using prime factorization technique? [CO3,C3, Mark: 8]

4. Determine in-degrees and out-degrees of all vertices of the following graph? [CO4,C2, Mark: 4]



5. Represent the following graph using an adjacency matrix? [CO4,C3, Mark: 8]





EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Final Examination, Fall 2020 Semester

Course: CSE106 Discrete Mathematics, Section-8
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time: 10 Minutes for uploading
Total Time: 1 Hour 20 Minutes

1/2



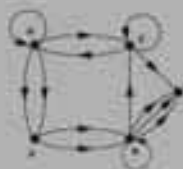
Note: There are Seven questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. Show that, $(x^3 + 2x)/(2x + 1) \text{ is } O(x^2)$. [CO3,C2, Mark: 8]
2. Determine the time complexity of the following algorithm. [CO3,C3, Mark: 8]

```

Input: A positive integer n.
Output: A integer (returned by the algorithm).
i ← 1
j ← 0
while (i < 2n) do
    i ← i + 1
return j
  
```

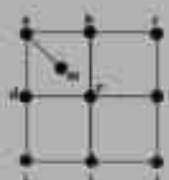
3. Calculate $\text{gcd}(910, 150)$ and $\text{lcm}(120, 240)$ using prime factorization technique. [CO3,C3, Mark: 8]
4. Determine in-degrees and out-degrees of all vertices of the following graph. [CO4,C2, Mark: 4]



5. Represent the following graph using an incidence matrix. [CO4,C3, Mark: 8]



6. Determine whether the following graph is a bipartite graph or not. [CO4,C3, Mark: 6]



September 14

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**EAST WEST UNIVERSITY****Department of Computer Science and Engineering****B.Sc. in Computer Science and Engineering Program****Final Examination, Summer 2021 Semester**

Course: CSE106 Discrete Mathematics, Section-4
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time 10 Minutes for uploading
Total Time 1 Hour 20 Minutes

Note: There are *Seven* questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. Show that, $(x^3 + 2x)/(2x + 1)$ is $O(x^2)$? [CO3,C2, Mark: 8]

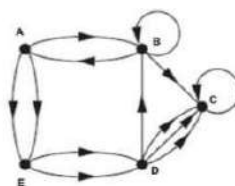
2. Determine the time complexity of the following algorithm as a function of n ? [CO3,C3, Mark: 8]

```

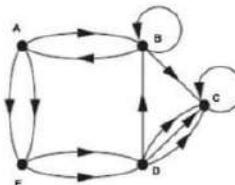
Input: A positive integer  $n$ .
Output: A integer (returned by the algorithm).
 $i = 1$ 
 $S = 0$ 
while ( $i \leq n^2$ ) do
    if  $i \% 2 \neq 0$  then  $S = S + i$ 
    if  $i \% 3 \neq 0$  then  $S = S - i$ 
     $i = i + 1$ 
return  $S$ 
  
```

3. Calculate $\gcd(910, 150)$ and $\text{lcm}(120, 240)$ using prime factorization technique? [CO3,C3, Mark: 8]

4. Determine in-degrees and out-degrees of all vertices of the following graph? [CO4,C2, Mark: 4]



5. Represent the following graph using an adjacency matrix? [CO4,C3, Mark: 8]



Course: CSE106 Discrete Mathematics, Section-7
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE Department
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes

Note: There are *SEVEN* questions, answer ALL of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

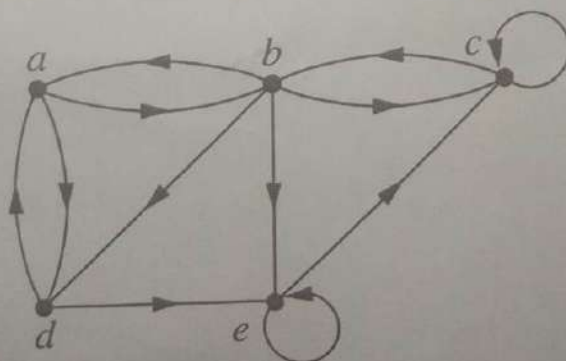
1. Show that the Big-O of, $f(x) = \frac{x^2 + 2x}{2x+1}$ is x^2 ? [CO3,C2, Mark: 8]
2. Determine the time complexity of the following algorithm as a function of n ? [CO3,C3, Mark: 8]

```

procedure matrix_man (n, resultnn: integers)
    for i = 1 to n do
        for j = 1 to n do
            resultij = i
        end_for
    end_for
    return resultij

```

3. Calculate gcd(9910, 1550) and lcm(1120, 2240) using prime factorization technique? [CO3,C3, Mark: 8]
4. Determine in-degrees and out-degrees of all vertices of the following graph? [CO4,C2, Mark: 4]



Course: CSE106 Discrete Mathematics, Section-7
Instructor: Dr. Mohammad Salah Uddin, Assistant Professor, CSE
Full Marks: 50 (25 will be counted for final grading)
Time: 1 Hour and 10 Minutes
Additional Time: 10 Minutes for uploading
Total Time: 1 Hour and 20 Minutes

Note: There are **Four** questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. **Show** that, (x^3) is $O(x^4)$ but, that (x^4) is not $O(x^3)$. [CO3
Mark

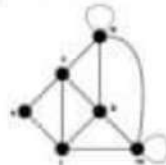
2. **Determine** the time complexity of the following algorithm. [CO3
Mark

```

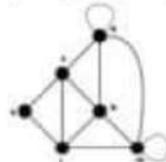
Input: A positive integer n.
Output: A integer (returned by the algorithm).
i = 1
S = 0
while (i ≤ n) do
    if i % 3 ≠ 0 then S = S + i
    i = i + 1
return S
    
```

3. **Calculate** gcd(120, 240) and lcm(120, 240) using prime factorization technique. [CO3
Mark

4. **Determine** degrees and neighborhoods of all vertices of the following graph. [CO4
Mark



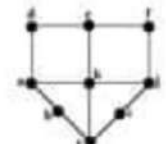
5. **Represent** the following graph using an adjacency matrix. [CO4
Mark



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January 10, 2021

6. **Determine** whether the following graph is a bipartite graph or not. [CO4
Mark



7. A full 3-ary tree has 3 internal vertices. **Determine** the number of vertices and number of leaves of that tree. [CO4
Mark